

Project Description

INSourced Training in Entrepreneurship for GRaduate Students In Translational Years -- INTEGRITY

INTEGRITY will place 141 STEM Ph.D. candidates inside research translational activities--social, commercial, local, international...anywhere research impacts the broader community--where they will be trained and encouraged to make innovative contributions and develop professional networks. Below we present (a) this innovation in graduate education; (b) its broader impacts; (c) how we will assess participant performance and evaluate the project; and (d) the results of prior NSF support on which this new project is in part based.

(a) Innovations In Graduate Education

Our pressing need

The US prepares STEM doctoral students well for jobs in academia, but not so well for careers outside academia.¹ Most of these students will end up in careers other than the tenure-track research positions for which they have been prepared²; the inefficiency³ and perhaps injustice⁴ of this system has been widely noticed. Nor has undergraduate life prepared them especially well: employers find recent grads as a whole quite unprepared.⁵ Grad students need a different kind of professional development to identify, aspire to, land and hold STEM jobs outside of academia, especially in areas of national priority. (Hence NRT⁶.) We propose to provide that needed professional development by *leveraging the connections universities already have outside of academia and preparing graduate students to add value to those connections* in ways that are useful both to those connections and to graduate students themselves. INTEGRITY will produce STEM Ph.D.s who are productive, innovative community-builders, and measurably so, based on evidence that will appeal to employers.

When the community we are building is a profession, this preparation for employment is professional preparation. The values which the professional community upholds are expressed in the habits and tendencies of its mature members, and these are upheld as virtues. Professional preparation is thus moral preparation, and the production of effective and innovative community builders is thus training in ethical leadership. Here, departing from widely accepted accounts⁷, we follow an ethics of virtue in the tradition of MacIntyre⁸. On his account, the standards internal to the integrated set of crafts as practiced by its mature members *are* the measure of virtue--there is no other external standard--in any community. In every community whose craft is the conduct of research, these community-building professionals will be well versed and committed to a widely-recognized standard of responsible conduct of research. In research communities which translate research into activity that impacts persons outside in yet broader communities, community-building professionals will recognize and advance their conduct in adherence to social, commercial and legal responsibilities. And the development of facility and leadership in the living out of those responsibilities--the virtues--requires interaction with members of those broader communities--professions, industry, clients, public--outside of the research community. But such interactions are generally not designed into our preparation of Ph.D.s in the STEM disciplines.

¹ US STEM Ph.D.s are not well prepared for careers outside of academia: __

² Most of these will need that preparation for such jobs: __

³ This lack of adequate preparation is bad for US competitiveness: __

⁴ It is also bad for the graduate students themselves: __

⁵ <https://www.insidehighered.com/news/2015/01/20/study-finds-big-gaps-between-student-and-employer-perceptions>

⁶ NRT is proposed as a partial remedy to this lack of preparedness: __

⁷ Many virtue theorists take virtue as a set of habits that promote adherence to some independent ethical standard: __

⁸ MacIntyre's account does not: the virtuous person is the measure, that is, there is no standard outside of the community of mature practitioners: __

Universities with large numbers of Ph.D.s *have* connections with communities outside of academia⁹: with industry groups in the R&D process; with investors and entrepreneurs in commercialization activity; with local, regional, national and international development and social service agencies in need of research guidance; and through their substantial network of alumni and other supporters who engage with universities to collaborate on the transformation of research into broader social good. We will call this complex interface the sphere of Research Translation (see Fig. 1, below.)¹⁰



In these research translational sectors, Universities employ cadres of professionals on the execution end of the research translational funnel, moving ideas to impact. Sometimes those professionals, and the members of communities outside of academia whom those professionals engage, are organized to connect with graduate students. There are degree programs that specialize at the research translational interface. The University of Notre Dame has several¹¹, including master's degrees in Patent Law; in Entrepreneurial Studies (ESTEEM); in Global Health; and soon, in Global

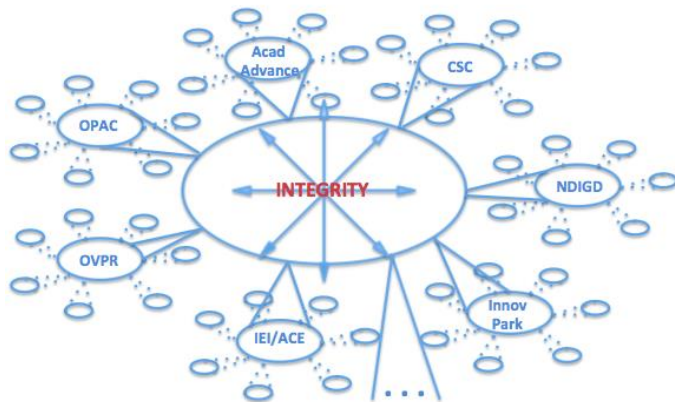
Affairs. Less formal, certificate programs offer for-credit clusters smaller than a separate degree program, such as the once-IGERT-funded GLOBES program that now resides in Notre Dame's Reilly Center for Science, Technology and Values. There are also single course-based programs, such as those encouraging service learning components with accompanying designations sponsored by our Center for Social Concerns; or single courses devoted to the translational enterprise, such as the Business On The Front Lines course in the Mendoza College of Business. These programs tend to be a mixture of theory, practice and reflection. But for all their advantages, these programs tend to be slower in response to targets of opportunity, limited in their capacity to scale, and have relatively high barriers to entry for community participants. We propose to add a new kind of offering to this mix.

The INTEGRITY innovation

Here's the core transformational strategy: add *one cross-cutting organizational change* to link graduate students with the research translational ecosystem just as we now do with research, but in a way that respects and enhances the primacy of research at Research (Carnegie classifications RU/H and VH) Universities.¹²

That change is the establishment of a consultancy inside the university, where information flows in on a voluntary basis from all research translational

Voluntary information flows in; voluntary assistance flows out



⁹ US universities lead the world in connectedness between research and industry, both in sheer number of connections and as a percentage of publications

¹⁰ The phrase "research translation" has been used this way in the literature, though it is more common in medical contexts than anywhere else.

¹¹ Provide URLs for the ND Institutions mentioned here

¹² Carnegie classifies research institutions as RU/H and RU/VH in this way

coordinating units, and efficient and targeted voluntary assistance from graduate students flows back out. (See figure 2.) Breath in, breath out: information in, assistance out. The fundamental idea is simple enough.

A consultancy?

We will establish a translational consultancy on campus. This is a place to come for assistance of the sort graduate students can provide to any research translational enterprise, at any phase of the research translational funnel. (See Figure 3.¹³) This consultancy will consist of graduate student Interns, including those who return in funded roles as Fellows on the leadership team; there they will be led by more experienced consultants from enFocus¹⁴, with the whole leadership team being mentored by professional consultants (from Accenture¹⁵: commitment still pending.)

Information flows in

To efficiently engage graduate students from all STEM (and eventually nonSTEM) disciplines, the flow of information and opportunities will have to be steady, reliable, and from at least a great many and perhaps all campus sources across the full spectrum of research translational activity. No such system currently exists, and INTEGRITY's first task will be to build one. (See Challenge 1, below.) It will have to play nicely with other information-gathering systems; happily many of these exist and can be tapped efficiently. The voluntary nature of the information exchange will support protection for private or otherwise sensitive information.

¹³ Applied knowledge has been represented through the metaphor of a funnel: http://ktdrr.org/ktlibrary/articles_pubs/ktmodels/

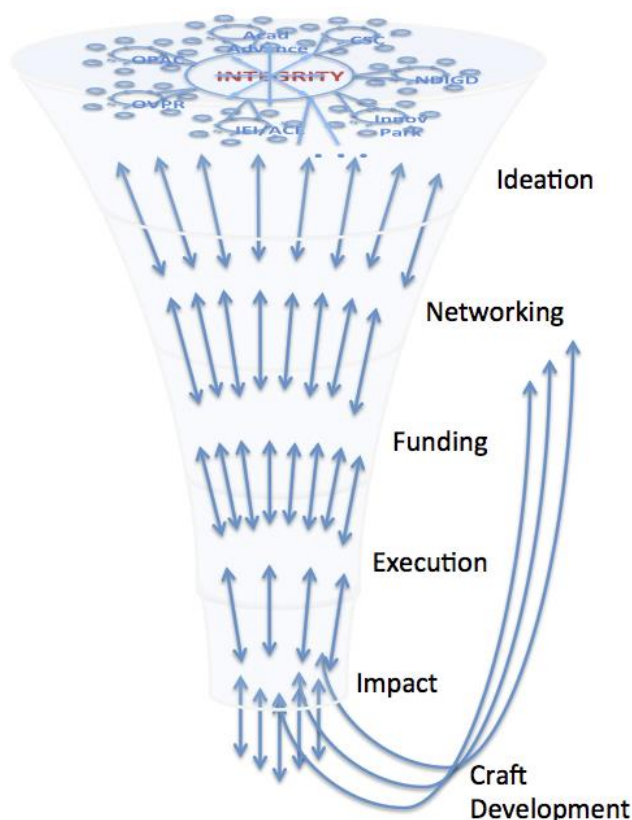
¹⁴ <http://sbenfocus.org>

¹⁵ Accenture is the world's largest consulting firm (as measured by revenue): <https://en.wikipedia.org/wiki/Accenture>

Assistance flows out

As information is gathered, opportunities for engagement will be identified and projects scoped. So as to add value to rather than compete with other campus professional development programs, opportunities will be funneled on a priority basis to graduate degree, certificate, and course programs when there is sufficient lead time. Projects suitable for nearer-term action will be sorted, scoped and posted on the INTEGRITY internal web site as manageably-sized practicums, offered on a competitive basis to any university graduate student. These practicums will be offered in both Internship and Fellowship varieties. Unpaid Interns in the basic program will have shorter (40 hr/semester) engagements. Some of these Interns may return on graduation for service (either volunteer or client-supported) as Translational Consultants. Longer engagements (10 - 15 hr/week) will be offered on a competitive basis to the most successful Interns, who will be named

Translational Year Fellows and supported at NSF Research Fellowship levels¹⁶ (of which the Fellow's lab will pay half, and INTEGRITY will pay half.) Projects at all levels of engagement--for Interns, Translational Consultants, and Translational Year Fellows--will be pre-scoped and aligned with graduate student skill sets, by members of an experienced leadership team who are themselves guided by professional consultants.



Piloting, Assessing, Sustaining, Replicating

The central challenges in building such a system are 1) coordinating the flow of information across university research translational units; 2) articulating a vision of maturity as members of the professional STEM community; 3) arranging for the attraction, preparation, uptake, and efficient engagement of graduate students in ways that produce in them the requisite sorts of professional development; and 4) launching the consultancy. Once the pilot program is built, remaining challenges include 5) tracking the impact of the system on graduate student careers, on research translational activity, and on university research; 6) sustaining the system; and 7) building and disseminating a replicable model, in accord with the University Mission and as an appropriate dividend on taxpayer investment. The INTEGRITY project will be presented in the course of responding to each of these challenges in turn.

Challenge 1: coordinating the flow of information.

There are many ways in which the University provides and tracks information: the Office of Development's network of Academic Advancement Directors; Notre Dame Research's Pre- and Post-

¹⁶ NSF Research Fellowship amount: http://www.nsf.gov/news/news_summ.jsp?cntn_id=134647

award departments, and their Office of Tech Transfer; the Center for Social Concern's office of Communications and Advancement; the Office of Public Affairs' Internal Communications and Public Affairs departments; and the EngageND system of archiving regional community impact. Notre Dame's Initiative for Global Development and our Innovation Park have robust communication systems, as does every College (Arts & Letters, Science, Engineering, Business, Architecture, Law, Global Affairs, and the Graduate School).¹⁷

All of these serve particular interests well, but none of them was designed to bring together information from all areas of research translational activity at every phase. We need a system to cut across and draw from any and all of these existing systems, as well as areas where no information is systematically gathered at all. We cannot collect all of the information, nor would we want to. We want only to collect that information which people working in each area are willing to share to attract graduate student assistance to their endeavors. The INTEGRITY information system will have two modes of intake: one consisting of active interviews, and the other of receiving regular reporting.

Surveying Mature Practitioners

In areas where information flow is known to be limited, we will engage graduate students to actively seek out the information needed. (The details this process and its advantage to graduate students are provided under Challenge 2, below.) We will offer this service to all university faculty. These already have some measure of assistance through ND Research's office of Pre-award and through the Directors of Academic Advancement working in Development and reporting to Deans and ultimately the Provost.¹⁸ But these systems do not have the resources to regularly engage individual members of the Faculty with creative assistance, particularly on the broader impacts/translational activity front. Providing graduate student assistance to each member of the faculty (or, rather, to any member on a voluntary basis) to create, populate, and keep updated a research translation and collaboration portfolio in a shared database will be an important outcome of the INTEGRITY program. Not only are the faculty very rich sources of ideas for graduate student engagement, but they are models whose experience and networks can be very useful examples to graduate students, whether they remain in academia or not. (In the pilot study, we would meet with over 100 faculty; as the program grows to scale at 10 TYFs and 200 Interns per year, we will be able to complete a pass of Intern meetings with all (at 50% interested) university faculty within three years, and hold Fellow-Intern meetings to review these portfolios and associated research translational projects at a rate of one Fellow-Intern meeting per week.)

The same process will be open for university staff who may not be members of the faculty, but who play pivotal roles in the research translational process; regular collaborators outside of the university (from industry, civic and governmental organizations) will also be included. Creating research translation and collaboration portfolios for these non-faculty members also will be a rich source of information and opportunity for graduate students. A monitoring and evaluation staff specialist in one of several university entities that employ them, for example, or a regional venture capitalist who regularly engages in University commercialization activities, would have valuable information to add both to the database and to young professionals looking to better understand research translational processes.

Partnering with Coordinating Units

In addition to these active attempts to gather research translational data from mature practitioners, INTEGRITY will set up a system to receive reports from the many campus units who already coordinate information flow through their own active efforts. Some of these efforts, such as the

¹⁷ Provide URLs for these ND units.

¹⁸ URLs for about ND Pre-Award, about Academic Advancement

network of Directors of Academic Advancement, operate near the ideation and networking end of the funnel; others, like the Office of Public Affairs regular gathering of 40 or more outreach professionals on campus, operate on the execution and impact side. During INTEGRITY's first year, we will invite five of these entities to contribute to the database: Academic Advancement (who connect to the chief academic officer--Deans and Center Directors--in every academic entity), STEM Community Engagement, the Office of Public Affairs (who run the EngageND database for regional impact), the Center for Social Concerns, and the Office of Tech Transfer. Others may be incorporated in subsequent years.

The Basic Unit

Whether developed from individual interviews or gathered and contributed from coordinating units, the basic unit of information flow will be the **Opportunity Report**: the identification of a proposed intervention, how it would enable which goals or objectives to be achieved, a financial measure of the value at stake, what resources are required, what skills from the proposed consultant, contact information for key collaborators in the activity, when the support would need to commence and how long would it take.¹⁹ These Opportunity Reports, one or two pages in length, will be posted to an internal INTEGRITY site (with ND login required for access.) INTEGRITY Interns and Fellows will be able to shop (browse, search, create custom search strings with notification) in this space for engagement opportunities that fit their skill sets and schedules. Two additional documents may be tied to each Opportunity Report: any Bid Document (consisting of a proposal to undertake a project, including a list of proposer's skill set match with campus reference contact information for each skill); and any Project Report (detailing project status for any successful bids.)

Here is an example of an Opportunity Report, and several more brief descriptions of kinds of projects suitable for INTEGRITY Interns and Fellows, follow:

Opportunity Report	
Name of Filer	FILERNAME
Date of Filing	5/1/2015
Opportunity Name	PROFNAME Engine Efficiency Optimization Research Nontechnical Summary
Short Description of Proposed Intervention	Read and summarize 3 of PROFNAME's papers in a format accessible to Corporate Social Responsibility (CSR) lead for engine manufacturer COMPANY, with follow-up response to edit requests from ND grant team members, COMPANY, or CONSULTANT representatives
Research Area Involved	Engine Optimization Software
Objectives and Goals to be Achieved	Secure generator donation from COMPANY as part of AGENCY grant submission by leveraging application of PROFNAME Engine Optimization research to new engine type, to motivate COMPANY participation in grant.
Financial Value (Range) of Most Likely Win	\$12000 - \$36000
● Justification	One to three large diesel generators donated contingent on grant success
Financial Value (Range) of Highest Value Win	\$1.5M to \$2M
● Justification	USAID grant reception may hinge on COMPANY participation
Resources Anticipated	3 papers identified by PROFNAME
Description of Skills Required	Ability to read and craft 1-2 page written summary of all three technical papers for a educated lay (CSR) audience, which might also be reviewed by COMPANY's technical team (estimated 3rd year Engineer or equivalent)
Skills Keywords	Engineering, Strong Writing, Team Player

¹⁹ Opportunity Reporting is the name given for this sort of activity in Salesforce and other CRM software: https://help.salesforce.com/apex/HTViewHelpDoc?id=reports_oppforesales.htm

Anticipated Collaborators	(each added separately by name and affiliation): ND Initiative for Global Development grant lead; ND Corporate Relations officer assigned to COMPANY, CONSULTANT Representative, COMPANY CSR Lead, AGENCY Program Officer
Contact Information if Outside University	(each added separately)
Earliest Start Date	Immediately
Latest Start Date	6/1/2015
Estimated Required Hours	10 - 20
Fee basis	volunteer or \$15/hr
Latest Completion Date	6/15/2015

This example is not fictitious; several partners in an important cluster of university-corporate relationships all had an interest in finding a way to collaborate around this opportunity. The need for a graduate student to play this role was explicitly discussed. There was no campus system to engage such assistance, however, and an opportunity for a graduate student to see into and contribute to a real-world problem, and develop a rich set of networking contacts, was lost. Such opportunities emerge regularly inside research translational arms at the University, and very often in ways that cut across seemingly very different areas. Consider three.

- Graduate students working in public-private partnerships--in areas as diverse as molecular diagnostics commercialization, disaster relief housing, or pre-college computer science education--will become known to the many professional consultants who work in such projects. Since these professional consultancies do not recruit Ph.D. candidates as part of their entering cohorts, but only to fill specific high-skill positions, contact and experience working with consultants in such partnerships would be a strong advantage for any applicant interested in this important class of employment opportunities.
- Readily engaging graduate students in commercialization efforts from any area of research is the first step to developing a farm system for young PIs, who are required for a significant commitment on SBIR or STTR grants.²⁰
- Much of the good work in K-12 STEM education that has been done in partnership with university programs such as GK-12 (our own included) could be facilitated piecemeal by Interns in multiple, shorter projects, or in exceptional cases by Translational Year Fellows working in K-12 education.

Formating the Data

All this data, gathered actively or passively, about individuals, units, collaborators, and priorities, will be contained, curated (with basic quality control standard operating checklist for intern-fellows meetings developed by enFocus fellows in year 1), stored, and made available for analysis. The form in which this data will be contained will have to change over time. The format needs to provide easy access to program participants; it needs to complement other data sources on campus; as it grows it will become more and more valuable and more appealing to other campus constituencies. In order to keep our upfront focus on the experience of new graduate students, our intention is to start with the basic containment structure that is already in place for everyone on campus: the network of apps in Google Drive (with Opportunity Reports entered in Forms, collected, analyzed, with auto-notifications driven by Apps Script in Sheets.) We will migrate to Salesforce, the world's leading Customer Relationship Management (CRM) software used by ND's Office of Development, in a pilot effort during year 2 and in full deployment during the third and final year of NSF support. The

²⁰ SBIR requirements: <https://www.sbir.gov/faqs/eligibility-requirements>

Opportunity Reporting begun in Drive is a central feature of CRM software and in Salesforce in particular. Google Spreadsheets import to Salesforce is supported by readily available software.²¹

Challenge 2: Articulating an account of maturity in the STEM Professional Community

In order for this system of information about research translational opportunities to be of benefit to graduate students, they must be able to add value to them. Graduate students have a great many of the skills required to participate effectively as force multipliers and innovators in these many varieties of research translational activity. But those skills should be identified, sharpened, and recorded. Working together with the graduate school professional development (PD) committee and inside the series of PD spires already present and guiding graduate student professional development at the University, we will prepare graduate students for translational consulting service by helping them to identify and achieve a common basic set of skills and outlooks.

A Sensible list of the desired skills is not difficult to formulate, so there are many²². Another, assembled by Kris Hamilton, founder of Koru, a firm dedicated to cultivating skills employers want: grit, polish, teamwork, curiosity, ownership, analytical rigor, and impact²³. Because we are building a system for the specific purpose of professional development of STEM Ph.D. candidates, we have formulated our own set of rubrics.

Rubrics for Professional Development

The **principles** guiding the creation of these rubrics are simple. First, they must be easily understood by busy graduate students; they must have an intrinsic appeal and point to an achievable upward trajectory; they must be clear enough to enable inter-rater reliability²⁴ (i.e., consistency in application of the rubrics) among untrained graduate students engaged in self- and peer-analysis; there must be enough challenge in the spectrum to separate the beginner and the advanced in statistically significant ways. Finally, the rubrics must be coherent with one another, and together present a whole which is appealing in its level of challenge to graduate students, impressive to prospective employers, and of sufficient size to be grasped as a whole by graduate students and prospective employers alike.

We assemble these areas of growth in three fundamental categories: **productivity, innovation, and community building**. An employee who is productive, innovative, and effective at contributing to a workplace, an organization, or the public good is on face value a good hire. Each of these three categories conveys some useful meaning, but can in turn be further analyzed.

Productivity is broken into three categories:: data analysis, project management, communication. Each of these categories is articulated in turn through a manageable rubric. *Data analysis*: from basic mechanical proficiency (compiling, sorting, cutting, plotting/visualizing, basic statistical analysis), to asking and answering meaningful questions with the one's own data, to asking and answering meaningful questions in a fresh context with new data, to effectively engaging in a productive public conversation around new data sets.²⁵ *Project management*: from managing individual tasks in terms of goals, objectives, and timelines; to negotiating simple group task in

²¹ <https://zapier.com/zapbook/google-sheets/salesforce/>

²² Here is one compilation: <http://www.asch.org/where-will-you-go-after-your-phdpostdoc/>, drawn from [[list sources here.]]

²³ https://community.asme.org/decision_point_dialogues/b/weblog/default.aspx accessed 12/22/2015.

²⁴ Inter-rater reliability: https://en.wikipedia.org/wiki/Inter-rater_reliability

²⁵ One useful overview of data science skills: <http://www.burtchworks.com/2014/11/17/must-have-skills-to-become-a-data-scientist/>; on the role of the citizen data scientist: http://www.lavastorm.com/assets/Citizen-Data-Scientist-Lavastorm_White_Paper.pdf

terms of goals, objectives, timelines, and responsibilities; adding identification and management of contingencies in individual and group contexts; to making effective use of basic project management software to track and communicate these fundamental project management tasks.²⁶ *Communication*: from making effective use of basic oral and written communication; to demonstrating also an awareness of audience and the ability to make effective rhetorical choices; to adding fluency and facility to the skill set; and finally displaying good judgment as to when to communicate, what--and what not--to communicate, and to whom.²⁷

Innovation is also broken into three categories: background searching, thinking from multiple perspectives, taking initiative. *Background searching*: from having the habit of asking the question of what has already been tried; to effectiveness at finding a reasonably complete set of answers to that question; to conducting these searches efficiently and collaboratively; to engaging in thoughtful analysis of past efforts, able to clean new insights from old attempts. *Thinking from multiple perspectives*: from taking the time to actually listen to what others think; to success in repeating back to others what they think to each other's satisfaction; to generating new insights that surprise and delight others, from their points of view; to effectiveness in seeking out a variety of useful alternative points of view. *Taking initiative*: from effective use of time in getting done what is expected; to distinguishing between the more important and the urgent, and making room for the more important; to sharpening up one's analysis of what is most important; to improve one's ability to systematically act in pursuit of what is most important.

Finally, **community building** consists of four categories: strategic networking, collaborative problem-solving, professional leadership, and respectful partnership development. *Strategic networking*: from intentionally cultivating professional connections; to being aware of the nature and value of strong and weak professional ties; to recognizing and filling gaps in one's set of connections; to efficiently curating a rich set of professional connections. *Collaborative problem-solving*: from working with others only when help is needed; to being aware of one's need for partners and a collaborative plan; to regularly working with others on such plans effectively; to effectively cultivating group dynamics in which all participants are empowered to add value in collaborative projects.²⁸ *Professional leadership*: from awareness of professional responsibilities/guidelines; to experienced practice of meeting responsibilities and operating within guidelines; to communal affirmation of professional responsibilities; to reflective public contribution to professional standards. *Respectful partnership development*: from recognition of the genuine interests of potential partners; to effective acknowledgment of those interests; to creative satisfaction of the genuine interests of multiple partners; to credible, demonstrable personal concern for such interests.

Each of these ten categories of achievement will be built out with descriptive examples, available online, highlighted in curricular options (such as the Social Responsibilities for Researchers for-credit option described below), and presented in voluntary professional development sessions offered by the INTEGRITY leadership team once each semester. Together, they are the marks of the sort of mature STEM Ph.D. holders that INTEGRITY is designed to produce and assist in transitioning into professional careers both in and outside of academia.

Challenge 3: Preparing and inviting graduate students

²⁶ Project management basics: <http://www.brighthubpm.com/project-planning/5582-getting-started-with-basic-project-management/> and <http://chronicle.com/blogs/profhacker/12-basic-principles-of-project-management/31421>

²⁷ Communication basics: here we have more modest aspirations than, but will learn from, the work that has been done in preparing graduate students for scientific communication, e.g., <http://research.fit.edu/instep/Communication.html>

²⁸ Drawn from ATC21S 1-dimensional empirical progression: <http://www.atc21s.org/emprirical-progressions.html>

Graduate students will vary widely as to where they fall along this spectrum of development. The professional development strategy we will follow in the integrity project is to a) facilitate graduate students' self assessment of their own skill sets across this spectrum, b) provide experiential opportunity to improve and/or display their developed skills; c) voluntarily invite peer- and mentor-affirmation of their own assessments; and d) present the skills as illustrated in publicly displayed projects, affirmed by peers and mentors participating in those projects. (This is fundamentally the LinkedIn strategy for professional representation, and we intend to leverage and improve upon that strategy.)

Graduate students will participate in the INTEGRITY program at one or more of three levels of engagement: that of the **Internship** (a 40-hour, unfunded introductory level), the **Translational Consultant** (*ad hoc*, volunteer or project-supported), and that of the **Translational Year Fellow** (a funded, much more substantial commitment.) All participants begin with the Internship program, which will consist of one 2-hour and two 4-hour introductory activities, a roughly 20-hour project, and two 5-hour capstone activities. Before these, a 2-hour Gateway activity will serve as an introduction to the program and cut across all these areas of professional development.

Six Professional Development Segments for Interns

1. That **Gateway activity** will function as a come-and-see invitation to the INTEGRITY program: required for entering, with an invitation to formally apply at the end of the 2-hour session. (During the pilot project, the importance of establishing a randomized control group post-application will also be explained.) The atmosphere will be one of a formal program offering of the Graduate School, with a welcome from the Dean or Associate Dean. The context of professional development in preparation for careers both in and outside of academia will be introduced. So also will the context of ethical leadership development, its relationship on our virtue ethics approach to career preparation, and the need from both perspectives for an account of professional maturity. In this context, the INTEGRITY program elements will be introduced, each of the common Internship activities by a representative of the host organization (all of this articulated, below.) Finally, the 10-category account of professional maturity will be covered, a time for reflection provided, and a baseline data set acquired. (To gather this data we will use a Google Form accessible from an easy URL in a bring-your-own-device context, with some additional portable workstations provided.)

Following this Gateway activity, a first cohort plus a randomized control group will be selected. The sole criteria are that each candidate is at least a rising third year ("Translational Years") graduate student in a STEM Ph.D. program in (Director-of-Graduate-Studies-affirmed) good standing, has attended the Gateway activity and submitted the baseline assessment. (Further detail on the study is discussed in Section (c), below.) After selection and notification, the INTEGRITY Internship program will commence. Training will take place in five additional activities. Four of these will take place in each of the Graduate School's already well-developed Integrated Professional Development framework, consisting of Ethics, Research, Teaching and Career Spires.²⁹

2. In the **Ethics Spire**, the Reilly Center for science technology and values already hosts a *Social Responsibilities of Researchers (SRR)* program for a cultivating ethical STEM leaders who are richly aware of the social contexts within which research takes place.³⁰ The Reilly Center's SRR program will offer a one-hour overview of the nature, extent and kinds of social impacts of research, with particular emphasis on the categories and examples offered in documentation of the Broader Impacts NSF review criterion. (We will introduce a template for reflection, mapping instances where the research itself, or some output directly related to it, or supported by and relevant to it,

²⁹ ND Graduate School Professional Development Spires: http://graduateschool.nd.edu/professional_development/vision-and-mission/

³⁰ Reilly Center SRR: <http://reilly.nd.edu/research/research-integrity/social-responsibilities-for-researchers/>

onto ten areas suggested in the National Science Board's Review and Revision of NSF Review Criteria.) This experience will culminate in a work product consisting of a research and reflection paper tracing the contours of the broader impacts in each participant's own area of research (or some subset approved by SRR staff.) The Reilly Center will offer a complimentary program in which new INTEGRITY translational consultants can select a *for-credit pathway* through the same INTEGRITY internship experience, where the core preparation and consultancy experience will be wrapped in further opportunities for reflection and development in a cohort context. Whether or not it is included in the for-credit component, this exercise will leave participants with a rich sense of the relevance of their own areas of research to a broad spectrum of social goods. Interns will be invited to generate Opportunity Reports from this reflection process.

3. In the **Research Spire**, the Graduate School already hosts the *Graduate Institute for Leadership Development (GILD)*, designed for participants to gain a better understanding of themselves, others, and the organizations in which they serve, and then to apply their training in a manner informed by the University's core values of integrity, accountability, teamwork, leadership in mission, and leadership in excellence."³¹ The GILD program will offer a one-hour orientation preparing Interns to **interview** one member of the faculty, staff, or key industry, community or government partners who are practitioners in the University's research and research translational activities. The purpose of these interviews will be to map (and as the program matures, to update) that practitioner's research interests, network of current and potential collaborators, and funding sources onto a rich spectrum of Broader Impacts possibilities connected with each area of research. From data collected first through online research and enriched through interviews with mature practitioners, graduate participants will generate a **Practitioner Profile** including a **Broader Impacts Survey** delivered through the Graduate School's Office of Grants and Fellowships (OGF). Though the full form is too long to include here, key categories are highlighted in the table below.

Practitioner Interview Categories									
Name of Interviewer	Date of Interview	Name of Practitioner	Research areas	Recent Grants Obtained Per Research Area	New Funding pursued	Key Collaborators	New Contacts Pursued	Metrics employed	Impact Documented
Broader Impacts Survey (Present, Anticipated, Possible)									
	Broadening Participation (for Women, Under-Represented Minorities, Disabled)	STEM Education (including Educator PD)	Public STEM Literacy and/or participation	Wellbeing of individuals	Workforce development and competitiveness	Partnership Development (academic, industry, others)	Economic Competitiveness - commercialization	National Security	Enhanced Infrastructure for Research & Education
Research itself									
Activities Directly Related									
Activities Supported and Complementary									

Each Profile will serve three purposes: 1) to better inform the OGF of the circumstances within which graduate students are conducting research, so as to better pair them with grant and fellowship opportunities; 2) to add to the INTEGRITY database of voluntary information, accessible to all contributing campus units; and 3) to provide a context in which graduate participants can generate or respond to Opportunity Reports for project completion.

4. In **20-hour Projects** undertaken within six months of these two introductory activities, students will gain practical experience working outside of their labs, and always in activities which involve persons outside the University. Projects are always responses to Opportunity Reports (ORs). Participants can choose ORs for their practicums in a variety of ways: 1) they may have generated those ORs, at the suggestion of the mature practitioner or at their own suggestion, in the context of

³¹ ND Ethical Leaders in STEM: http://graduateschool.nd.edu/professional_development/ethics/ethical-leaders/

reviewing a practitioner's existing network of opportunities; 2) they may have generated those ORs in some other area, informed and inspired by their review of the broader impacts of their own research (under the SRR exercise) or by opportunities that occurred to them on close inspection of a practitioner's network of research translational activity; 3) they may search the online database of pre-scoped ORs and compete for one or more of these. However they come by them, any OR chosen by a participant for their Internship Project will need to have its scope reviewed and approved by the INTEGRITY leadership team. (The composition of that team will be detailed under Challenge 4: Launching the Consultancy--below.) As part of the process of review by the participant and the INTEGRITY leadership team, ORs may be generated for later action and contributed to a database that will grow increasingly rich with time; future Interns may shop for projects in this space as well.

5. Once participants have completed Internship level projects at the 20-hour level, they will file a Project Report (roughly parallel in structure to Opportunity Reports) that will set the stage for a second pair of common activities. In the **Teaching Spire**, the Kaneb Center for teaching and learning already hosts a variety of opportunities for graduate students to increase their public speaking skills.³² The Kaneb Center will host the first of two required exit activities: a 1-hour session for the entire cohort will review best practices for public presentation; two-four weeks later, participants will gather in groups of roughly a half-dozen to make 10-minute presentations based on their completed projects. The target audience level will be members of the general educated public: taxpayers with interest in how our federal research dollars are spent, and with what Broader Impacts. (In the case of projects with predominantly corporate funding, the target audience will be stockholders, and the language will include that of corporate social responsibility.)

6. Finally, in the **Career spire**, the Graduate School's office of Grad Career Programs already encourages a variety of modes of contact with the broader Notre Dame alumni and collaborator networks, including the use of LinkedIn.³³ The final common required activity will be the creation of a project writeup in Notre Dame's online portfolio infrastructure. In addition to a brief project description crafted with eventual placement on LinkedIn in mind, the portfolio version will include articulation of the skills demonstrated on the project, both in terms of the very same rubrics used for each candidate's initial self-evaluation, as well as for skills listed as required on the approved Opportunity Report. (We will attempt to refine the craft of skills identification as the INTEGRITY project advances.) Moreover, peers and mentors who also worked on the project will be invited to *endorse* those skill claims as demonstrated on each project, and to *recommend* the Intern based on *that project*. If the Intern judges it to be in his/her career interest, this internal online document can be elevated to public status, with endorsements, recommendations, and project descriptions moved to the Intern's LinkedIn profile. In such LinkedIn profiles, skill endorsements, recommendations and project descriptions will be linked in a way not currently supported by LinkedIn, by hyperlinking out from the LinkedIn project description to the ND online portfolio description.

Training Fellows

Once their projects are posted to the online portfolio (whether or not they have selected the option to post them to LinkedIn), Interns are named Translational Consultants. They are now eligible to both file and bid on additional Opportunity Reports, some of which may carry modest stipends from the submitting entity. (The first Internship project must be undertaken on a volunteer basis; subsequent projects need not be.) Finally, Translational Consultants are eligible to apply for Translational Year Fellowships (TYFs).

³² Kaneb Center Programs: <http://kaneb.nd.edu/programs/>

³³ Notre Dame Graduate Career Program: http://graduateschool.nd.edu/professional_development/career/

Because they will have first been Interns and then Translational Consultants, , Translational Year Fellows will have already received the training described above. As top performers in this space, they will have demonstrated the aptitude to lead. They will receive on-the-job training as part of the leadership team with the PI, one or more enFocus consultants (on whom see more under Challenge 4, below), and participation of career consultants (from Accenture*: not yet confirmed.)

Challenge 4: Launching the Consultancy

INTEGRITY needs to hit the ground with a leadership team, eventually composed largely of graduate students. We need experience to begin: to identify and scope projects with and for interns, to establish a pattern of credible force multiplication from the consultancy among campus constituents; to assist in building the database and dashboard. We will import that experience from recent graduates of the university, who have gained the necessary experience in curricular and practical contexts through the ESTEEM³⁴ and enFocus programs. ESTEEM is a Notre Dame master's degree program in STEM entrepreneurship; enFocus is a consultancy (hiring largely ESTEEM grads) offering insourced consulting services to business, government, and social agencies in Michiana. enFocus currently runs a successful Internship and Fellowship program, in part supported by a Lilly Foundation grant through the University of Notre Dame.

Our schedule of program implementation begins in September 2016 and moves through increasingly shorter cycles with increasingly larger cohorts.

	Leadership Team												Total onboarded Interns
	PI	enFocus Fellows	Fellows	Interns									
	Fall				Spring				Summer				
Year 1	1	3	0	0	1	3	0	36	1	3	0	36	36
Year 2	1	2	4	36	1	2	4	45	1	2	4	45	45
Year 3	1	1	5	45	1	1	5	60	1	1	5	60	60
	Sept	Oct	Nov	Dec	Jan	Feb	March	April	May	June	July	Aug	141
	EnFocus Fellows Begin	Internship Experience Prefigured	Information Pipeline Flowing	Intern Invitations Prepared									
2016	Projects I Fellows I Onboarded	Projects I	Projects I	Projects I	Interns Invited	Orientation Prepared	GateWay I	SSR I	Interviews I	Projects I	Projects I Fellows I Apply	Projects I Fellows I Invited	
2017	Projects II Fellows II Onboarded	Projects II	Projects II	Projects II	Projects I	Presentation I, Post I	Gateway II	SSR II	Interviews II	Projects II	Projects II Fellows II Apply	Projects II Fellows II Invited	
2018													
2019					Presentation II, Post II	Gateway III	SSR III	Interviews III	Projects III	Projects III	Projects III (Fellows III Apply*)	Presentation III, Post III (Fellows III Invited*)	(* Beyond the term of the NSF Award)

Challenge 5: Tracking the Impact

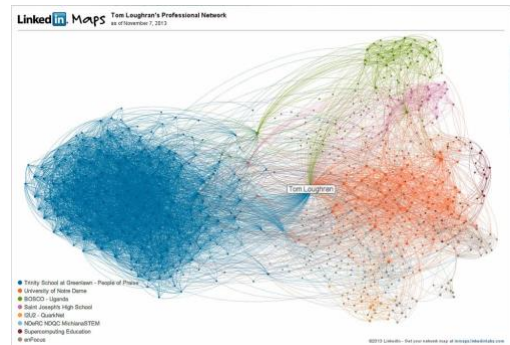
Beyond the impact on graduate students which will be studied under NSF funding and described in section (c)--Performance Assessment/Project Evaluation--below, INTEGRITY will track the impact of its activities on the growth of the University's entire network of research translational activity. Using social network analysis and data available both internally and on LinkedIn, we can model networks, track the growth of community, and intervene strategically based upon the findings of network science. One such finding is that job growth comes through weak ties, and during the Posting activity led by Grad Career Programs, graduate student participants will be encouraged in the use of software

³⁴ ESTEEM: <http://esteem.nd.edu/about-master-of-entrepreneurship-program/>

such as Socilabs³⁵ and Gephi³⁶ to analyze their own networks and to develop ties in industries they would like to explore as career options.

Challenge 6: Sustaining the System

In order for any consultancy to succeed, it must add value to its clients. The primary client of the INTEGRITY consultancy is the University. INTEGRITY, like any program, will have a net cost associated with it, and a net return for clients, and thus an ROI (Return on Investment). We will calculate costs of the program based upon actual and opportunity costs. Return will be estimated by net savings or income to clients--from projects, from value due to increased benefits to graduate students, and from an increase in grant funding due to better Broader Impacts review marks--multiplied through by a percentage of credit assigned by clients (including individuals, academic units, and ND Research) for INTEGRITY participation. A net ROI will be calculated that can be compared to other University programs; INTEGRITY will have to hold its own in that comparison. As INTEGRITY succeeds against that set of internal university metrics, we will begin to raise external funding for Named Translational Year Fellowships (and eventually, Postdocs.) We will also pursue corporate sponsorship for elements of the program. Both of these strategies will reduce program cost and thus improve ROI.



Finally, INTEGRITY is expected to be housed in the upcoming Idea Center (IC.) In the IC, professional champions will be employed to advance commercial and social innovations proposed by faculty and identified as priorities by chief academic officers through our Office of Development's network of Academic Advancement Directors. INTEGRITY will bring to the IC system three additional advantages: a) the energy and innovation from graduate student participants; b) a "bottom up" flow of ideas from the implementation end of the research translational funnel, to complement the "top down" flow from university researchers; and c) a supporting infrastructure for Broader Impacts, not only advancing research translational/broader impact activities directly, but indirectly moving the University toward greater competitiveness for NSF Center-level funding through development of sustained and coordinated metrics for broader impacts of research.

Challenge 7: Disseminating Results and Replicating Success

The results from our three-year pilot project of INTEGRITY – both the study of undergraduate student employability measures, and what we learn from managing and tracking the growth of the research translational network – will be written up for publication and presented at professional meetings. (Add to this paragraph from the corresponding section in the CCE stem proposal, *ceteris paribus*.)

The INTEGRITY model has an additional pathway for dissemination which is not typically available to university programs. In addition to publishing as much of the standard operating procedures for setting up a program like integrity as we can capture in that format, we will have a group of professionals in the enFocus consultancy who will be both experienced in setting up such a program, and available for hire to assist in replicating it at other universities. The enFocus business model

³⁵ <http://socilab.com/#about>

³⁶ <https://gephi.org/users/>

already includes packaging their own Internship/Fellowship model, and replicating it, starting with other Indiana cities. Already in 2016, and focus will be replicating its current model for the time in Elkhart, Indiana. The Lilly Foundation has expressed a strong interest in seeing programs like that already succeeding in enFocus replicated across the state. It has already devoted considerable resources in partnership with all major Indiana Universities to create such internship and fellowship programs in their surrounding areas, as part of a statewide effort to slow the flight of top talent that is produced by Indiana's many excellent universities.³⁷

INTEGRITY provides the possibility of such a program being built not only for recent STEM graduates, but also for current STEM Ph.D.s and master students. INTEGRITY's collaboration with enFocus may be the first partnership in the nation attempting to build together complementary internship/fellowship programs designed first to equip graduate students for employment while they are at the University, and then for retaining a number of them in regional jobs after they matriculate. That enFocus has made the promulgation of such programs part of its business model bodes well for our ability to scale and actually offer sustainable assistance in replicating the INTEGRITY model at other universities nationally.

(b) Broader Impacts

Not only in its instance at Notre Dame but as emulated elsewhere, the INTEGRITY approach will systematically introduce a very broad range of STEM Ph.D.s to the idea of thinking creatively about Broader Impacts, had to call them with the venue for making a contribution at a formative stage in their careers. Many of these will stay in academia, and will be better prepared to broaden the impacts of their own research for having participated in the INTEGRITY program. Still more Ph.D.s, even at Notre Dame and certainly on the national average, will move onto careers outside of academia, either immediately or eventually on completing their degrees. These new doctoral recipients will be far more keenly aware of the networks in which universities participate with a very kinds of industries, government, and social entities in which most new Ph.D.s will find positions. They will have seen at a much younger point in their careers than would otherwise be the case how persons in their new sets of responsibilities can reach back to connect their own activities with universities in a broad variety of research translational activities.

Current university faculty, also, will have the opportunity to think systematically about the broader impacts of their own areas of research during the INTEGRITY process, and to participate in conversation with fresh younger minds in thinking creatively of new opportunities for translating their research into Broader Impacts. Moreover, the wisdom forged by the efforts of many university faculty will become more transparent in a system which archives every effort and encourages replication of every success. Collaborations between potentially similar areas of Broader Impact activity from different faculty members will be more likely to occur when all that activity is systematically documented, subject to creative enhancement in conversations between faculty members and rising STEM Ph.D.s, and rendered transparent to a team of INTEGRITY leaders who dedicate substantial time to thinking creatively and collaboratively about Broader impacts.

Data is a wonderful thing. The ability to archive data about an ever-expanding variety of research translational activity opens up a great many avenues for mining it productively. In the INTEGRITY program, groups at the University who specialize in broadening participation in particular areas--encouraging more women in STEM, or more Hispanic Americans in mathematics, or more African-

³⁷ Lilly Endowment gift to ND for enFocus: <http://news.nd.edu/news/45260-notre-dame-receives-3-million-lilly-endowment-grant-to-support-enfocus-program/>

American male engineers--will be empowered by tailoring the data set with keywords particularly useful for their diversification efforts. Given the centralized opportunity we will have at the Reilly Center for training those gathering or coordinating the gathering of information, we can and will include sensitization to the use of language in the documentation which renders it usefully searchable by a broad range of constituencies, including those many already on campus whose interests will be communicated during the training process.

Finally, enabling longitudinal study and ensuring consistent use of common metrics across a variety of different programs are important but often elusive targets of any university broader impacts infrastructure. The INTEGRITY approach supports cross-cutting transparency and long term archiving of data, both of which are necessary components of any longitudinal and statistically sizable studies, as has been pointed out by the National Science Board in its Revision and Review of NSF's Merit Criteria.³⁸

(c) Performance Assessment/Project Evaluation

To track the impact of the system on graduate student careers, on research translational activity, and on University research, we will investigate the impact of the internship experience, further translational consultant activity, and the translation year fellowships. We will use two measures of graduate student employability: their growth along 10 professional development lines, as measured by self, peer, and mentor assessments; and their degree of connectedness, as measured by connections, endorsements, recommendations, and project links from their LinkedIn profiles. (We will also track employment using the Graduate School's Academic iNDEX³⁹, though these results will be collected beyond the term of NSF funding.)

Integrity will measure two sets of states before the internship program: a) each candidate's self evaluation across 10 areas of professional development (at the time of the Gateway activity for all interested applicants); and b) each candidate's LinkedIn contacts (at the time of applicant acceptance.) At the end of each Internship (at the Career-posting event), post-Internship Translational Consultant activity (on an annual basis), and at the end of any Translational Year Fellowships activity (at an Exit Interview), both of these measures will be taken again. In addition to the self-assessment of each candidate in each of the 10 categories of professional development, we will also seek peer and mentor evaluation at every measurement after the baseline.

A randomly selected control group of eight will be set up at the same time as the initial cohort of Interns, who will report the same professional development self-assessment, and their LinkedIn statistics. The same group will be followed for one year. After that time and in fairness to these randomly selected members of each control group, they will be allowed to participate as Interns with their progress tracked, but not as part of the new cohort. (This will reduce Cohort II's statistical size to 37, and Cohort III's to 52.) Also, a randomly selected group of eight will be selected from among the first year's Translational Consultants and function as a control against which to measure the impact of the Translational Year Fellowships. Evaluation of the program will be conducted by personnel from Notre Dame's Center for Social Research.⁴⁰

Evaluation surveys will be applied at the end of each of INTEGRITY's six training activities for formative assessment. Results will be discussed within the month with recommendations for program modification by the Leadership Group (PI, CoPI(s), enFocus Fellows and TYFs. These

³⁸ <http://www.nsf.gov/nsb/publications/2011/meritreviewcriteria.pdf>

³⁹ https://graduateschool.nd.edu/assets/145127/academic_index_flyer.pdf

⁴⁰ ND's Center for Social Research: <http://csr.nd.edu/>

surveys and proposed modifications will be reviewed between introductory and exit activities by an Advisory Board consisting of representatives from each participating coordinating unit (Academic Advancement, Center for Social Concerns, the Office of Public Affairs and Communications, the Office of Tech Transfer, and the Coordinator of STEM Community Engagement). Modifications recommended will be approved by the Leadership Group, discussed with each Cohort during one or both of the exit training activities, and incorporated into the following year's procedure.

(d) Results from Prior NSF Support

Dr. Laura Carlson is an internationally recognized cognitive scientist working in the area of spatial cognition. She is a Professor of Psychology and the Dean of the Graduate School, as well as Vice President and Associate Provost of the University. In 2009, Carlson was appointed Associate Dean of Professional Development in the Graduate School, and over the next four years led a team of professionals from across the university to create our current robust and comprehensive Professional Development program. In July 2013, Carlson was promoted to Dean of the Graduate School. Carlson has extensive administrative experience, and has trained graduate students and postdoctoral scholars who have moved on to a variety of placements in the academy and industry. Her research focuses on spatial cognition, examining how spatial representations are influenced by core cognitive processes of attention, memory, perception, and language; thus, the research projects in her lab fall at the intersection of spatial cognition and any of these other areas, leading to a broad and multi-faceted approach that includes empirical, computational, cross-linguistic, and psychophysiological methods. As Co-PI, Carlson will manage program assessment and provide oversight for the entire project. Most relevant to the INTEGRITY project is Carlson's current CCE-STEM project, Cultivating Leaders for Ethical STEM (NSF Award #1449469, from 1/1/15 - 12/31/18, \$410,340, training 93 STEM Ph.D.s)

Dr. Thomas Loughran is a philosopher in the Physics Department at the University of Notre Dame, a social entrepreneur in an academic setting. A specialist in the Broader Impacts of STEM research, Loughran has worked extensively with leading national education and public outreach organizations in particle physics (QuarkNet), astronomy (NOAO's TLRBSE and NASA's Spitzer Teacher and NITARP), computer science (the SCxx conference series Education Program); and nanotechnology (the NISE NNetwork). As Education Program Leader for I2U2 at Fermilab and later at Notre Dame, Tom led educational design and training for online research environments introducing high school students to data from CMS and LIGO-Hanford. For the past four years he has worked closely with consultants from Accenture on a variety of projects, one of which--rural electrification and entrepreneurial training in sub-Saharan Africa--involved grants also from HP, Lenovo and SunEdison. He is a founding member of the Board of Directors for enFocus. As PI, Loughran will coordinate the Leadership Team and coordinate all Internship Training with the Graduate School.

Loughran spearheaded, wrote, and served as managing co-PI for the Notre Dame extended Research Community (NSF Award # 0638723, from 7/1/07 - 6/30/13, \$2,710,000.) In many respects, NDeRC served as a forerunner to INTEGRITY, focusing only on broader impacts in K-12 education; 19 STEM Ph.D.s completed 37 graduate-student years of research and education apprenticeship. The public report required by NSF is available.⁴¹

Intellectual Merit

⁴¹ <http://tinyurl.com/ndercpublicreport>

Over its six-year history, NDeRC funded the research of 19 graduate students, each for two years, in Physics, Chemistry, Biological Sciences, Electrical and Mechanical Engineering, Computer Science, and Sociology. In addition, these 19 Fellows (14 of whom have so far completed their Ph.D.s) were effectively apprenticed as education and public outreach specialists, with particular emphasis on our Integrated STEM Community model. On that model, STEM education is best understood as effective invitation into STEM community, whether on a track into specialized STEM research, into participation in its commercial and social applications, or in every case into a STEM-literate humanity to whom both responsibility for STEM research and its benefits rightfully belong. Fellows gained experience with the tools we use to invite teachers and students into the STEM research community: with our Annual Forum (attracting 360 K-12 teachers over six years, almost half to more than one), with a range of social media tools (including more than 750 blog posts), and with our Institute/STEMweek model. That model, which invites K-12 educators into week-long immersions into the language and practice of various areas of STEM research and collaborates with them on a discipline-specific, week-long classroom experience for their students, has now been extended into genetics and developmental biology, nanotechnology, environmental science, astronomy, bacterial transformation, and particle physics.

Broader Impacts

Over these six years, NDeRC STEMweeks have been in 500+ different classrooms with 124 Michiana teachers and over 17,000 students in 52 different schools. Our flagship STEMweek program, BioEYES, brought hands-on genetics studies with zebrafish to some 14,500 students. A survey of 332 middle and high school girls showed the percentage who reported that science was fun increased by a third, while the number who found science frustrating or boring was cut roughly in half, after their week-long BioEYES experience. Nearly all (>97% of) participating teachers report being well prepared in their Institute week for the classroom activities, and they report consistently that their Institute/STEMweek participation positively encouraged multiple best practices in other areas of their teaching.

References

INTEGRITY References (Still in progress)

1. US STEM Ph.D.s are not well prepared for careers outside of academia:
2. Most of these will need that preparation for such jobs:
3. This lack of adequate preparation is bad for US competitiveness:
4. It is also bad for the graduate students themselves:
5. Nor does undergraduate education prepare them for these jobs:
6. NRT is proposed as a partial remedy to this lack of preparedness:
7. Many virtue theorists take virtue as a set of habits that promote adherence to some independent ethical standard:
8. MacIntyre's account does not: the virtuous person is the measure, that is, there is no standard outside of the community of mature practitioners:
9. US universities lead the world in connectedness between research and industry, both in sheer number of connections and as a percentage of publications:
10. The phrase "research translation" has been used this way in the literature, though it is more common in medical contexts than anywhere else:
11. Provide URLs for the ND Institutions mentioned here.
12. Carnegie classifies research institutions as RU/H and RU/VH in this way:
13. Applied knowledge has been represented through the metaphor of a funnel:
http://ktddr.org/ktlibrary/articles_pubs/ktmodels/
14. enFocus URL: <http://sbenfocus.org>

15. Accenture is the world's largest consulting firm (as measured by revenue): <https://en.wikipedia.org/wiki/Accenture>.
16. NSF Research Fellowship amount: http://www.nsf.gov/news/news_summ.jsp?cntn_id=134647
17. Provide URLs for these ND units.
18. URLs for about ND Pre-Award, about Academic Advancement
19. Opportunity Reporting is the name given for this sort of activity in Salesforce and other CRM software: https://help.salesforce.com/apex/HTViewHelpDoc?id=reports_oppforesales.htm
20. SBIR requirements: <https://www.sbir.gov/faqs/eligibility-requirements>
21. Sheets-to-Salesforce conversion:
22. Compilation of required skill sets for Ph.D. candidates on the nonacademic market:
23. Koru founder set of required skills:
24. Inter-rater reliability: https://en.wikipedia.org/wiki/Inter-rater_reliability
25. One useful overview of data science skills: <http://www.burtchworks.com/2014/11/17/must-have-skills-to-become-a-data-scientist/>; on the role of the citizen data scientist: http://www.lavastorm.com/assets/Citizen-Data-Scientist-Lavastorm_White_Paper.pdf
26. Project management basics: <http://www.brighthubpm.com/project-planning/5582-getting-started-with-basic-project-management/> and <http://chronicle.com/blogs/profhacker/12-basic-principles-of-project-management/31421>
27. Communication basics: here we have more modest aspirations than, but will learn from, the work that has been done in preparing graduate students for scientific communication, e.g., <http://research.fit.edu/instep/Communication.html>
28. Drawn from ATC21S 1-dimensional empirical progression: <http://www.atc21s.org/emprirical-progressions.html>
29. ND Graduate School Professional Development Spires: http://graduateschool.nd.edu/professional_development/vision-and-mission/
30. Reilly Center SRR: <http://reilly.nd.edu/research/research-integrity/social-responsibilities-for-researchers/>
31. ND Ethical Leaders in STEM: http://graduateschool.nd.edu/professional_development/ethics/ethical-leaders/
32. Kaneb Center Programs: <http://kaneb.nd.edu/programs/>
33. Notre Dame Graduate Career Program: http://graduateschool.nd.edu/professional_development/career/
34. ESTEEM: <http://esteem.nd.edu/about-master-of-entrepreneurship-program/>
35. Sociolab:
36. Gephi:
37. Lilly Endowment gift to ND for enFocus: <http://news.nd.edu/news/45260-notre-dame-receives-3-million-lilly-endowment-grant-to-support-enfocus-program/>
38. NSF Merit Review Criteria
39. ND's Academic iNDEX
40. ND's Center for Social Research: <http://csr.nd.edu/>
41. NDeRC Public Report

NDeRC Scientific and Educational products:

Achievement Type	Citation
Educational product - Institutional presentation Conference	*Antonelli, L. J.; A Summary of the Particle Physics Collaboration's activities: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum V; February 2012; University of Notre Dame; Notre Dame, IN.

presentation/poster	
Scientific product Conference presentation/poster	*Antonelli, L. J.; Beyond Standard Model Higgs Decaying to Tau Pairs at CMS. Presented at DPF 2011, Providence, RI.
Educational product - Institutional presentation Conference presentation/poster	+Beebe, A; A Summary of the BioEYES Collaboration's activities: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum V; February 2012; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional presentation Conference presentation/poster	+Brandt S, *Crocker M, *Goss V, *Quardokus R, +Rose L; An Invitation to the 2010 NANO Summer Institute: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum IV; January 2011; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional presentation Conference presentation/poster	+Brandt, S., +DelleDonne, K., *Wasio, N; A Summary of the NANO Collaboration's activities: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum V; February 2012; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional presentation Conference presentation/poster	*Carr, A., +Gensic, J., *Raycroft, F.; A Summary of the GENO Collaboration's activities: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum V; February 2012; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional presentation Conference presentation/poster	*Carr A, *Raycroft F, *Slabbekoorn A; An Invitation to the 2010 GENO Summer Institute: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum IV; January 2011; University of Notre Dame; Notre Dame, IN.
Scientific product Conference presentation/poster	*Crocker M, X. Sharon Hu, and Michael Niemier, "Design and Comparison of NML Systolic Architectures" International Symposium on Nanoscale Architectures, pages 29-34, June 2010 Oral presentation given June 2010 in Anaheim, California, USA
Educational product - Institutional presentation Conference Presentation/poster	+DelleDonne, K., *Dolan, M., +Fletcher, C., +McNeely, A.; A Summary of the ASTRO Collaboration's activities: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum V; February 2012; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional presentation Conference presentation/poster	*Dolan M, +Fletcher C, +McNeeley A, *Rueff K; An Invitation to the 2010 ASTRO Summer Institute: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum IV; January 2011; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional	+Gensic J, *Mikels J, *Rodak C, +Walsh M; An Invitation to the 2010 ENVIRO Summer Institute: A presentation to local K-12 STEM

presentation Conference presentation/poster	Teachers at the Collaborating for Education and Research Forum IV; January 2011; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional presentation Conference Presentation/poster	Loughran, T. (June 12, 2012) From 0 to 60: A Roadmap for becoming a designer of online learning environments. Reimagining K-20 STEM Education Conference at the University of Notre Dame.
Educational product - Institutional presentation Conference Presentation/poster	*Mikels, J., +Streeter, K.; A Summary of the ENVIRO Collaboration's activities: A presentation to local K-12 STEM Teachers at the Collaborating for Education and Research Forum V; February 2012; University of Notre Dame; Notre Dame, IN.
Educational product - Institutional presentation Conference Presentation/poster	Mooney, P.; Summary of NDeRC's activities presented to the following departments at the Univ. of Notre Dame: Chemistry and Biochemistry, Civil Engineering and Geological Sciences, Computer Science and Engineering, Mechanical Engineering and Physics; (Spring 2012).
Scientific product Conference presentation/poster	*Quardokus, R.C., Yuhui, L., Wasio, N.A., Guo, S., Kandel, S.A., (2010) Scanning Tunneling Microscopy Study Comparing Neutral and Mixed-Valence Dinuclear Organometallic Molecules on Au(111). 84 th Colloid and Surface Science Symposium. University of Akron, Akron, OH, USA.
Scientific product Conference presentation/poster	*Raycroft, F., Hyde, D. (2011) The Role of Socs Proteins in Regulating Light-induced Retinal Regeneration in Zebrafish. Midwest Zebrafish Conference. Mayo Clinic, Rochester, MN.
Scientific product Conference presentation/poster	*Raycroft, F., Kassen, D., and Hyde, D. Role of Socs Proteins in the Light-damaged Regenerating Zebrafish Retina. Poster Presentation at 9 th International Zebrafish Meeting. June 2010. University of Wisconsin. Madison, Wisconsin.
Scientific product Conference presentation/poster	*Rodak, C. and Silliman, S. (2010) "Application of Probabilistic Risk Analysis and Fault Trees to Wellhead Protection". American Geophysical Union Fall Meeting, San Francisco, CA. Dec 16, 2010.
Educational product - Institutional presentation Conference Presentation/poster	+Streeter, Kathe, +Knapp, Koren and *Mikels-Carrasco, Jessica. (2012) Placing SEEDS: A teacher/researcher collaboration. The University of Notre Dame, Graduate Student Union Research Symposium, poster presentation.
Educational product - Institutional presentation Conference proceeding	+Streeter, Kathe and *Mikels-Carrasco, Jessica. (2012 - submitted) Placing SEEDS: Science and ecology education development series at a public primary Montessori academy. Abstract submitted for American Montessori Society 2013 annual conference.
Educational product - Instructional product Curriculums and lesson plans	+Brandt S, *Crocker M, *Goss V, *Quardokus R, +Rose L; NANOweek: Lesson Plans for K-12 Classrooms

Educational product - Instructional product Curriculums and lesson plans	*Carr A, Peyer D, Peyer J, *Raycroft F, *Slabbekoorn A; GENOweek: Lesson Plans for K-12 Classrooms
Educational product - Instructional product Curriculums and lesson plans	*Dolan M, +Fletcher C, +McNeeley A, *Rueff K; ASTROweek: Lesson Plans for K-12 Classrooms
Educational product - Instructional product Curriculums and lesson plans	+Gensic J, *Mikels J, *Rodak C, +Walsh M; ENVIROweek: Lesson Plans for K-12 Classrooms
Educational product - Instructional product Curriculums and lesson plans	+Gensic J, *Mikels J, *Rodak C, +Walsh M; 2010 ENVIRO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Instructional materials	+Rose, L. (Summer 2011) Biology Modeling Workshop. University of Notre Dame.
Educational product - Instructional product Instructional materials	+Rose, L. (Summer 2011) Chemistry Modeling Workshop. University of Notre Dame.
Educational product - Instructional product Instructional materials	*Rueff, K., *Dolan, M., +Fletcher, C., +McNeeley A., +Modlin, S., +Havens, M., +Dove, C., +Chance, R.,: Digital Visualization Theater Presentation of A Journey Through the Solar System & the Universe.
Scientific product Journals - Non-refereed	Chatrchyan, S., *Antonelli, L. J., et al. [CMS Collaboration], Measurement of the Inclusive Z Cross Section via Decays to Tau Pairs in pp Collisions at 7 TeV, arXiv:1104.1617
Scientific product Journals - Non-refereed	Chatrchyan, S., *Antonelli, L. J., et al. [CMS Collaboration], Search for Neutral MSSM Higgs Bosons Decaying to Tau Pairs in pp Collisions at 7 TeV, arXiv:1104.1619
Educational product - Educational research Journals - Non-refereed	*Quardokus, R. C., *Wasio, N. A., Kandel, S. A.; (2012) The Scanning Theremin Microscope: To be submitted to The Physics Teacher.
Scientific product Journals - Refereed	Gupta, K., Joyce, M., *Slabbekoorn, A., Zhu, Z., Paulson, B., & Goodson, H. (2010). Probing Interactions Between CLIP-170, EB1, and Microtubules. J Mol Biol 395(5) pp.1049-1062.
Scientific product Journals - Refereed	Lu, Y., *Quardokus, R., Lent, C.S., Justaud, F., Lapinte, C., Kandel, S.A., (2010) Charge Localization in Isolated Mixed-Valence Complexes: An STM and Theoretical Study. J. Am. Chem. Soc., 2010, 132, 13519-13525. doi:10.1021/ja105958p.
Educational product - Educational research Journals - Refereed	*Mikels-Carrasco, Jessica. 2010. Nature in our own backyards: Urban ecology and children. Children, Youth and Environments 20(2): 190-199.
Scientific product Journals - Refereed	*Mikels-Carraso, Jessica. (2012 - forthcoming) Sherwood Washburn's New Physical Anthropology: Rejecting the "religion of taxonomy." History and Philosophy of the Life Sciences. 34.

Scientific product Journals - Refereed	*Quardokus, R.; Lu, Y.; *Wasio, N.; Lent, C.; Justaud, F.; Lapinte, C.; Kandel, S. (2012) Through-Bond Versus Through-Space Coupling in Mixed-Valence Molecules: Observation of Electron Localization at the Single-Molecule Scale. J. Am. Chem. Soc. 134(3): DOI: 10.1021/ja208981y.
Scientific product Journals - Refereed	*Rodak, C. and Silliman, S. (2011) "Probabilistic risk analysis and fault trees: Initial discussion of application to identification of risk at a wellhead". Advances in Water Resources xxx (2011) xxx-xxx. doi:10.1016/j.advwatres. 2011.02.005 Article in press.
Educational product - Institutional presentation Seminar	Loughran, T. (June - December 2011) CMS e-Lab Workshops for Teachers, presented to the QuarkNet Centers at Idaho State, University of Illinois-Chicago, University of Minnesota, University of Puerto Rico, Kansas University.
Educational product - Institutional presentation Seminar	*Raycroft, F., Hyde, D. (2012) The Role of TNF-alpha and the Stat3 Pathway during Light-induced Retinal Regeneration in Zebrafish. University of Notre Dame, Cell Seminar.
Educational product - Instructional product Teaching modules	+Beebe A, +Moran J; 2011 BioEYES Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Teaching modules	+Brandt S, *Crocker M, *Goss V, *Quardokus R, +Rose L ; 2010 NANO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Teaching modules	+Brandt S, *Goss V, *Quardokus R, *Wasio N ; 2011 NANO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Teaching modules	+Brandt, S; +Rose, L; *Goss, V; Loughran, T. Bringing Nanotechnology to K-12 Classrooms: The NANOweek Project (April 1, 2012). The NSTA 2012 Annual Meeting, Indianapolis.
Educational product - Instructional product Teaching modules	*Carr A, +Gensic J, *Raycroft F; 2011 GENO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Teaching modules	*Carr A, Peyer D, Peyer J, *Raycroft F, *Slabbekoorn A; 2010 GENO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Teaching modules	*Dolan M, +Fletcher C, +McNeeley A, *Rueff K; 2010 ASTRO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Teaching modules	*Dolan M, +Fletcher C, +McNeeley A, *Rueff K; 2011 ASTRO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers
Educational product - Instructional product Teaching modules	*Mikels J, *Rodak C, +Streeter K; 2011 ENVIRO Summer Institute at the University of Notre Dame: Professional Development for K-12 STEM Teachers

